

Data Visualisation

Week 2 – Perception and Categorisation

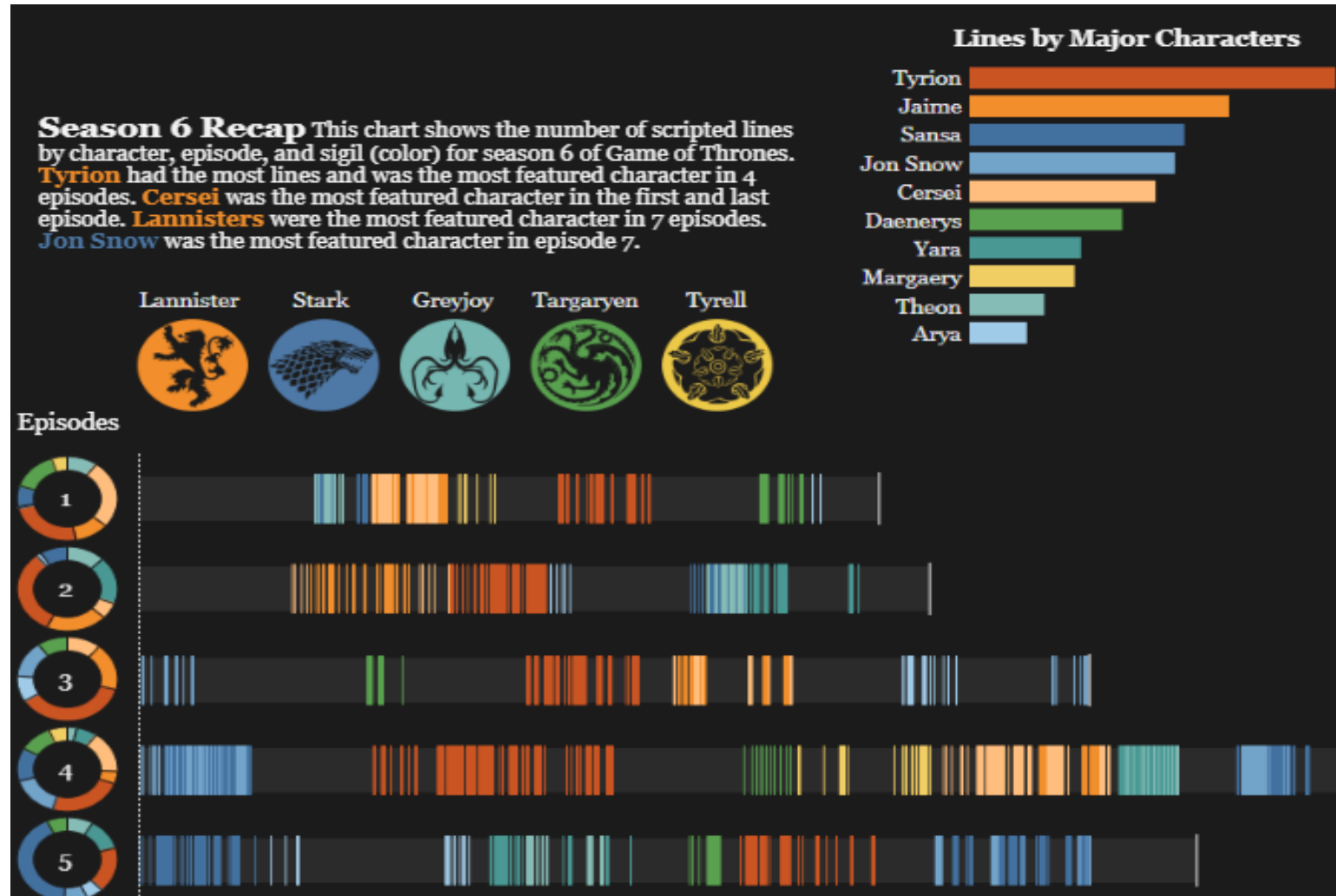
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Learning Outcomes Lecture 2

- Demonstrate understanding of how humans perceive the world around them on a general level and absorb complex data/information on a specific level.
- Analyse and evaluate how metaphors are used to convey unfamiliar information.
- Analyse and evaluate how mental models aid in the interpretation of complex visual displays.
- Select, formulate and integrate metaphors to suit data-driven tasks
- Design effective visualisations based on principles from perceptual psychology, cognitive science, graphic design and visual art.
- Demonstrate an understanding how visualisation is used in data journalism to communicate complex ideas and stories
- Demonstrate understanding how visualisation is used in story telling

Visualisation of the Week



<https://public.tableau.com/app/profile/adam.e.mccann/viz/GameofThronesSeason6Review/GOTS6Recap>

UnVisualisation of the Week



Tufte advocates a
minimalist perspective

High data to ink ratio.

UnVisualisation of the Week

MONSTROUS COSTS

Total House and Senate
campaign expenditures,
in millions



Recommended reading:

Useful Junk? The Effects of Visual Embellishment on Comprehension and Memorability of Charts

Scott Bateman, Regan L. Mandryk, Carl Gutwin, Aaron Genest, David McDine, Christopher Brooks

<http://hci.usask.ca/uploads/173-pap0297-bateman.pdf>



Overview

- Human Perception and memory
- How to categorise visualisations
- Class Exercises

Memory and Human Perception

Remember!

Well-designed visual representations can replace cognitive calculations with **simple perceptual inferences** and **improve comprehension, memory, and decision making**

By making data more **accessible** and **appealing**, visual representations may also help engage more diverse audiences in exploration and analysis

The challenge is to create **effective and engaging visualisations** that are **appropriate** to the data

Human Memory and Visualisation

- The power of visualisation is the ability to get across lot more complex information than what our memory can generally hold
- There are 3 types of memories that process information in our brain-
 1. Iconic memory or Sensory memory
 2. Working memory
 3. Long term memory

Human Memory and Visualisation

Type of Memory	Iconic Memory	Working Memory	Long Term Memory
Duration	1 second approx.	1 minute approx.	1 second – lifetime
Processing	Pre-attentive processing, before we pay attention	Hold and process about 5-9 chunks of information (Miller's Law)	Information stored through rehearsal or repeated application

- Long term memory is where things we memorize or remember are stored
- Iconic and working memories are the ones that interact with visualisations, so let's look at them in depth

Iconic Memory

- When we see a visual, the information remains in the iconic memory for less than a second. We process and store information automatically in this time
- This process is called **pre-attentive processing** and it happens automatically, before we pay attention to the information
- The pre-attentive process detects several visual attributes

Understanding how to make a particular attribute stand out can help us create visuals that emphasize the more important information

Working Memory

- The memory in use when working with a visual
 - Sensory information of interest to us is processed in the working memory
 - Information stays here for about a minute
 - Capacity of our working memory is between 5 to 9 similar items (Miller's Law)
- The capacity of our working memory can be increased by a process called **chunking**
 - Grouping similar items together

Working Memory

- Data visualisations take advantage of chunking
- When information is displayed in visuals that show meaningful patterns, more information can be chunked together
 - When we look at a visual, we can process much more information than when looking at the data in a table
- For a visualisation to be effective, we must not provide more data than what our brains can process
 - It is also important to display the visual on a single location, so we can see it without having to scroll or bounce back and forth between multiple locations

Human Perception for Visualisation?

- In order for our visualisations to be useful, it is important that they are easily interpreted by a viewer
- Bad data visualisations can be difficult to read and understand
- To visualize data effectively, it is useful to follow design principles
 - Derived from an understanding of human perception
- In order to create effective visualisations we must understand perception

What is Human Visual Perception?

- Vision is the first component of the human sensorial system and it is acquired fastest
- Our vision system has parallelism capacity
 - while it focuses on one point, the visual system can reach a wide surrounding area
- The human sensorial system is stimulated by a continuous flow of events
 - Resulting in a neural excitement called **sensation**
 - This process is extremely **fast and efficient**
- This continuous flow of sensations generates what it is known as perception
- Perceptions are different according to the characteristics of the stimulus
 - They are **interpreted**, allowing the brain to extract knowledge
 - The act of thinking, processing and making comparisons is known as **cognition** and is **much slower and less efficient**

Human Visual Perception and Data Visualisation

- Data visualisation is an external aid to support our working memory
- Visual perception is selective
 - If we tune our awareness to everything, we will be overwhelmed. So we selectively pay attention to things that catch our attention
- Our eyes are drawn to familiar patterns
 - We see what we expect to see. So visualisation must take into account what people know and expect
- Our working memory is very limited
 - We can hold a very limited amount of information in our memory when looking at a visual
- We want more visual perception and less cognition

Gestalt Theory

“It is not possible to have knowledge of the whole through the parts, but of the parts through the whole; the whole is more than the sum of its parts”

- Through the perception of the totality, the brain can realize, decode and assimilate an image or a concept
- It proposes a number of laws that which explain the structural and functional principles of the perceptive field:

Some Laws of Gestalt Theory

- **Proximity**: Objects that are close together are perceived as a group
- **Similarity**: Objects that share similar attributes (e.g., colour or shape) are perceived as a group
- **Enclosure**: Objects that appear to have a boundary around them (e.g., formed by a line or area of common colour) are perceived as a group
- **Closure**: Open structures are perceived as closed, complete, and regular whenever there is a way that they can be reasonably interpreted as such
- **Continuity**: Objects that are aligned together or appear to be a continuation of one another are perceived as a group
- **Connection**: Objects that are connected (e.g., by a line) are perceived as a group

Categorisation of Visualisations

Short Glossary of Terms

- **Chart:** Something that shows qualitative information (e.g., flow charts)
- **Data Dimension:** One single channel of data (e.g. a stock graph may comprise dates and prices - each is a unique dimension)
- **Designer:** The creator of a visualisation

Short Glossary of Terms

- **Encoding:** The visual property applied to a dimension of data that encodes the information into a visual medium for decoding by the reader
- **Graph:** Something that shows quantitative information (e.g., pie graphs and bar graphs)
- **Infographic:** visualisations that are manually generated around specific data, tend to be data-shallow, and are often aesthetically rich

Short Glossary of Terms

- **Reader:** The consumer of a visualisation, often someone other than the designer. The reader has information needs that are meant to be satisfied by the visualisation
- **Visual Property:** A characteristic that you can see (e.g. colour, size, location, thickness, and line weight)
- **Variability of a data dimension:** The values that are present or allowed in a single data dimension (e.g. integers vary discretely; position can vary continuously; categories are finite (and discrete, though may be hierarchical); numbers are infinite)

Remember!

Well-designed visual representations can replace cognitive calculations with **simple perceptual inferences** and **improve comprehension, memory, and decision making**

By making data more **accessible** and **appealing**, visual representations may also help engage more diverse audiences in exploration and analysis

The challenge is to create **effective and engaging visualisations** that are **appropriate** to the data

Classifications of visualisations

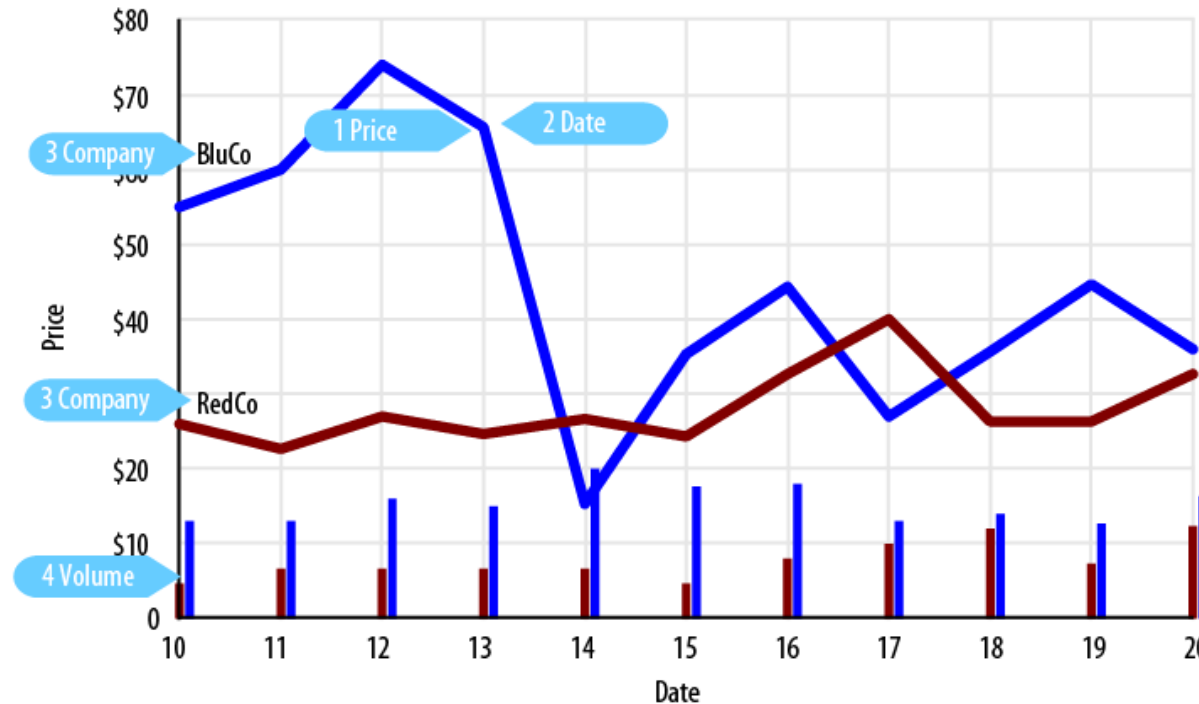
1. Complexity
2. Infographics vs Data visualisation
3. Exploration vs Explanation
4. Informative vs Persuasive vs Visual Art

Classifications of visualisations

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Complexity

- Classify by counting how many different *data dimensions* are represented
- The number of discrete types of information that are visually encoded in a diagram

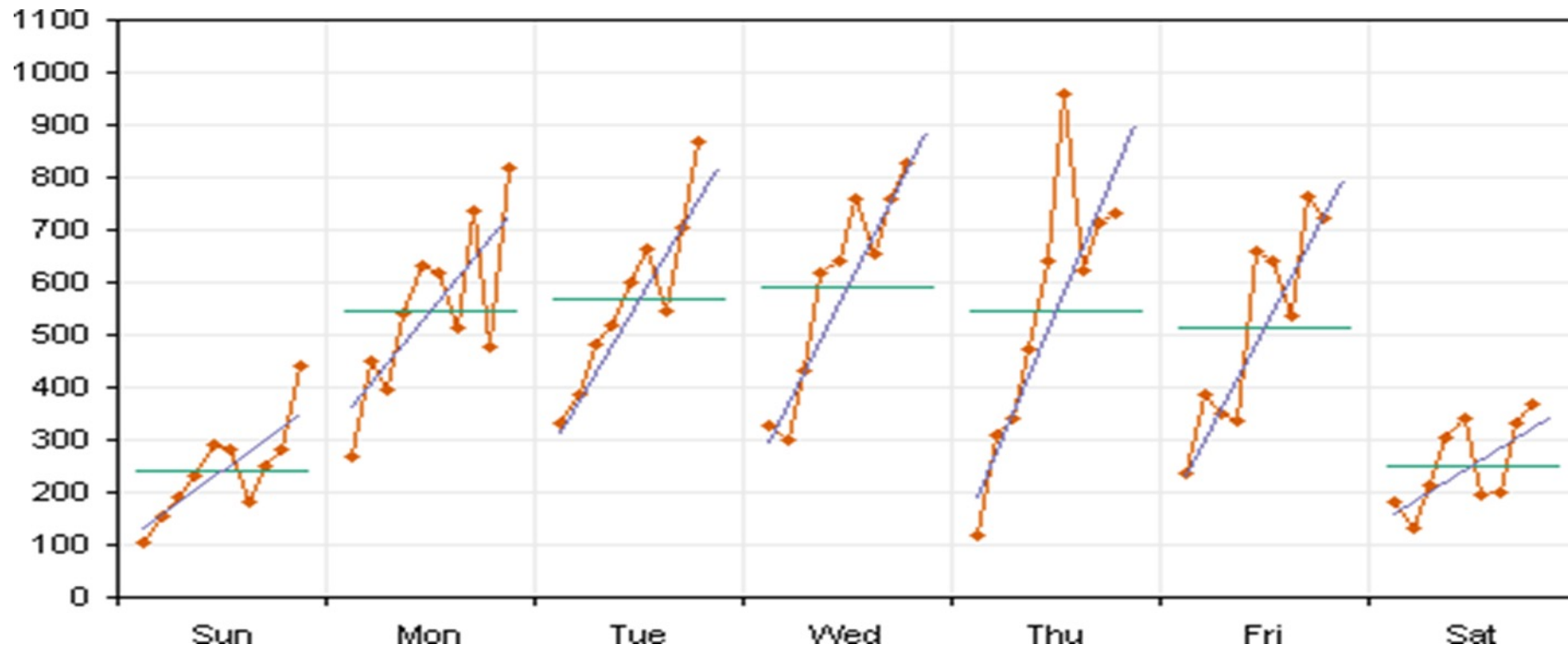


Complexity

- For example, a simple line graph may show the price of a financial instrument on different days: 2 data dimensions



Complexity



Cycle Plot – Days of the Week mapped over 10 weeks

Complexity

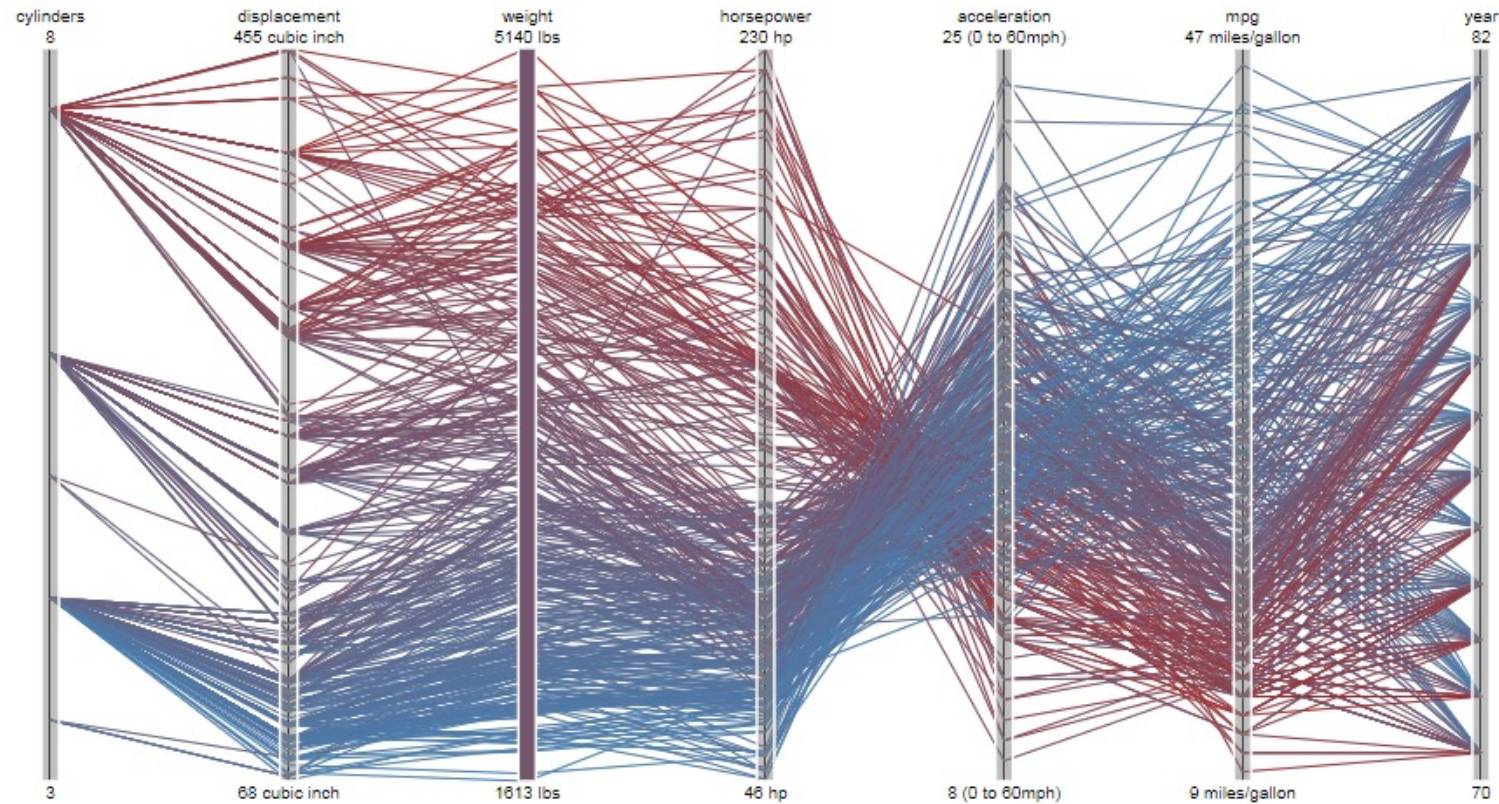
- As visualisations become more complex:
 - They are more challenging to design well
 - Can be more difficult to learn from
- For that reason, visualisations with no more than 3 or 4 dimensions of data are the most common
 - Visualisations with six, seven, or more dimensions can be found but may be more difficult to read!

Complexity

- 2 main challenges to designing more complex visualisations:
 - The more dimensions you need to encode visually, the more individual visual properties you need to use
 - There are relatively few well-known conventions, metaphors, defaults, and best practices to rely on
 - There is more of a burden on the designer to make good choices that can be easily understood by the reader
 - However, there are smart ways to think about and represent the data

Complexity

Parallel Coordinates of Automobile Data



A database of cars is plotted in seven coordinate dimensions; each path represents one car. Drag and resize the coordinate selection sliders to filter the cars in any dimension.

Source: GGobi

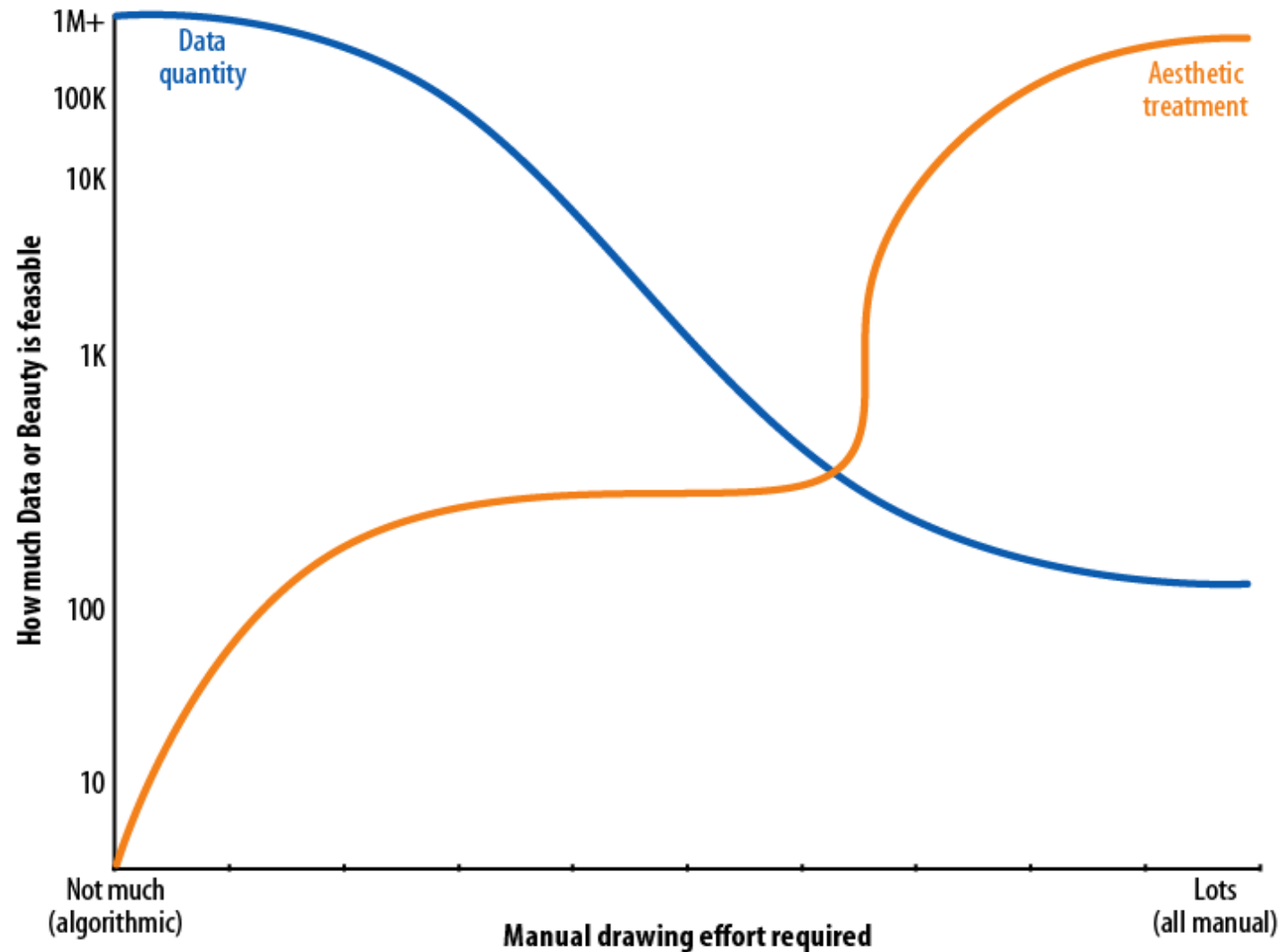
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Infographics vs Data visualisation

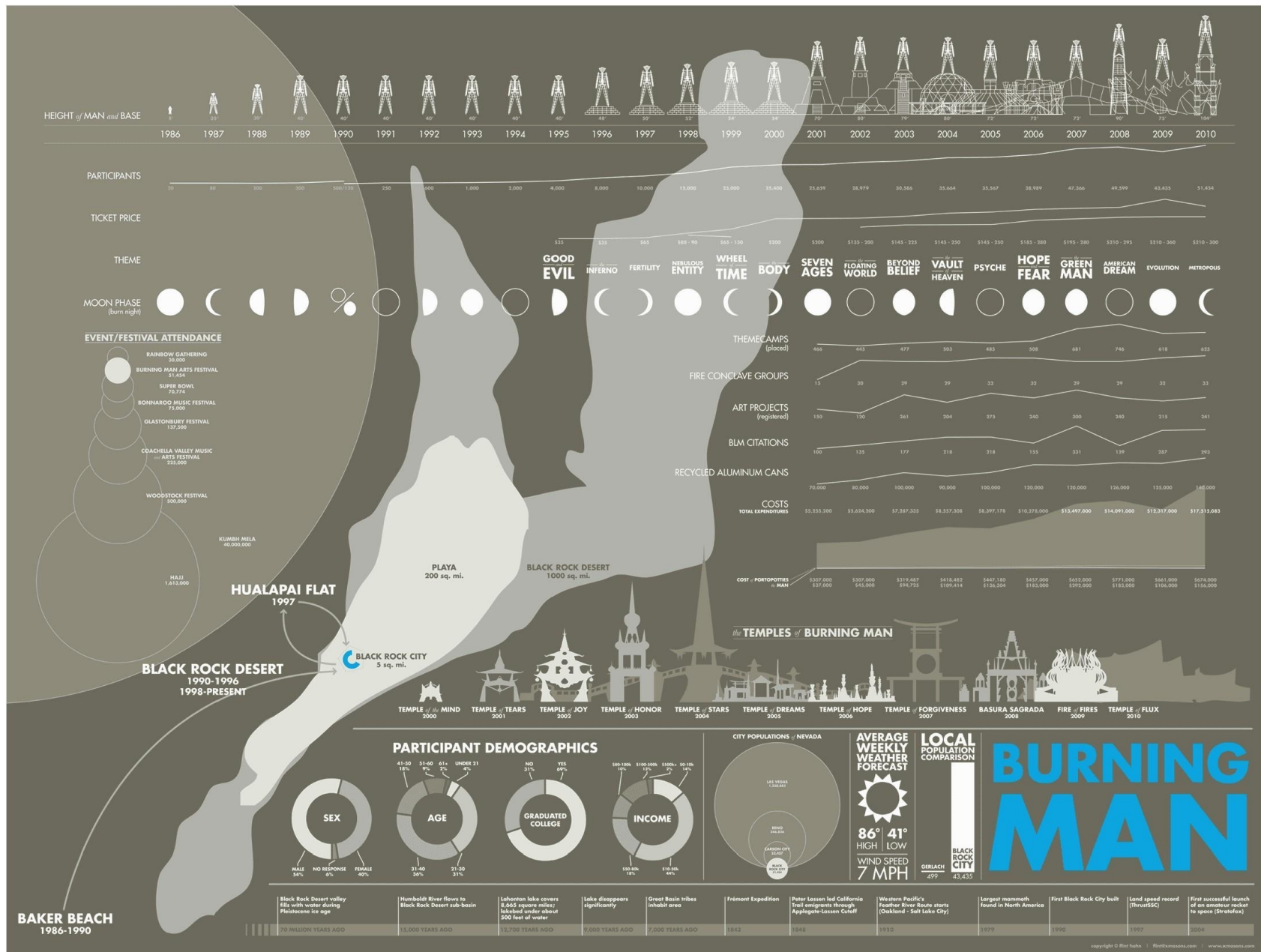
- **Infographics** and **data visualisation** are slightly nebulous terms often used interchangeably, but a distinction is worthwhile
 - **Infographic** used to refer to representations of information perceived as casual, funny, or frivolous
 - **Data visualisation** used to refer to designs perceived to be more serious, rigorous, academic or scientific!

Infographics vs Data visualisation



Infographics

- Iliinsky & Steele suggest that the term infographics is useful for referring to visual representation of data that is:
 - Manually drawn (and therefore a custom treatment of the information)
 - Specific to the data at hand (and therefore non-trivial to recreate with different data)
 - Aesthetically rich (strong visual content meant to draw the eye and hold interest)
 - Relatively data-poor (because each piece of information must be manually encoded)



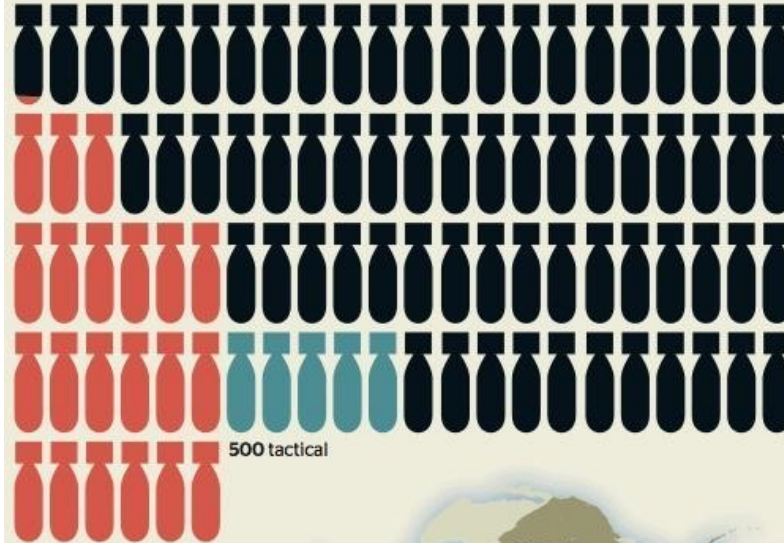
United States 9,400 total warheads

Heading downwards

World's most powerful and reliable arsenal. Some 200 warheads deployed in Europe; additional 2,500

warheads are spares and not counted as operational. Years of balanced reductions with the Russians in both strategic and tactical

warheads. Committed under President Obama to new talks to limit each side to 1,000 strategic warheads



2,126 strategic

France 300 total warheads



300 strategic

Stable A recently renewed arsenal. Small reduction announced in 2008 to less than 300 warheads, but no pressing need for change or updates. Strong political attachment to independent deterrent

China 240 total warheads



180 strategic

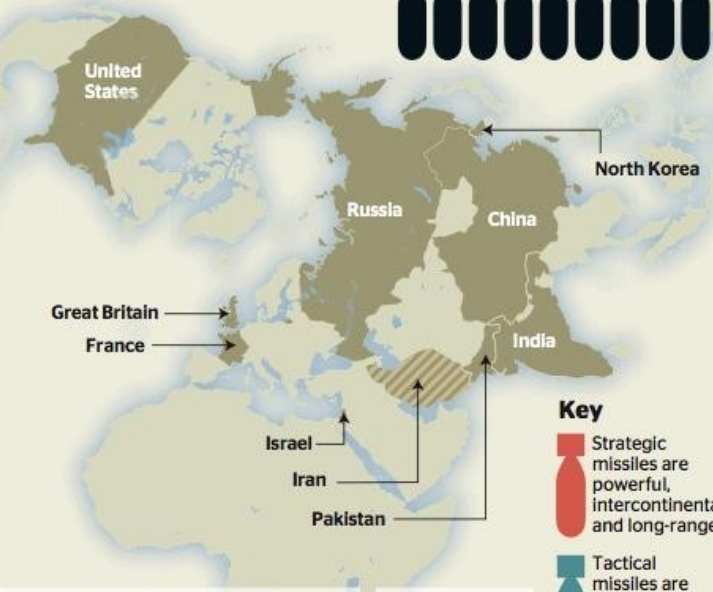
Stable Strong commitment to enough weapons to deter attack on its heartland, but no attempt at parity with the US or Russia. Many "strategic" weapons for regional use

UK 185 total warheads



160 strategic

Stable Despite talk of including small arsenal in multilateral talks, no firm commitment to do so. One submarine probably will be cut, but no guaranteed change in total of 160-185 warheads



Key

Strategic missiles are powerful, intercontinental and long-range

Tactical missiles are short-range and battlefield

Weapons in store or awaiting dismantling

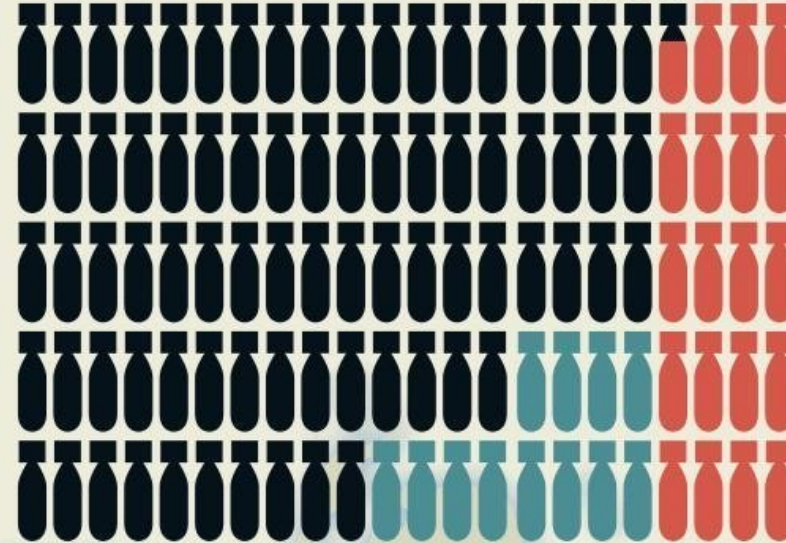
Source: FAS

Russia 13,000 total warheads

Heading downwards

Big reduction from 1991 total of 15,000 warheads. A quarter of present total probably awaiting

dismantling. An ageing arsenal, expensive to maintain and update. Every incentive for further deep cuts



2,050 tactical

2,668 strategic

Pakistan 70-90 total warheads



60 strategic

India 60-80 total warheads



60 strategic

Slowly increasing Approximate equilibrium and no new nuclear arms race likely. All details and number are secret. Only strategic, not tactical missiles, held by each side

North Korea Fewer than 10 total warheads



Fewer than 10 strategic

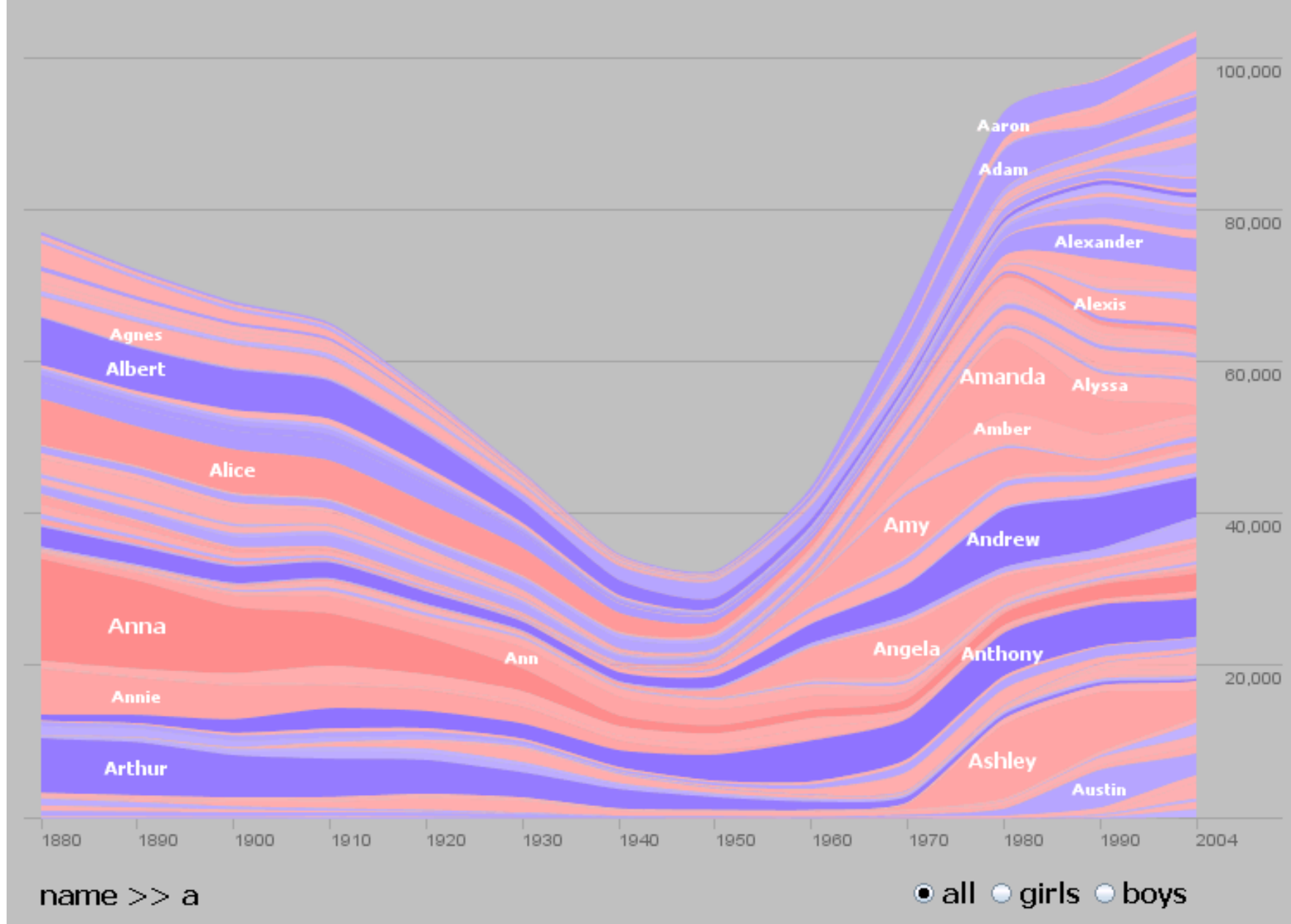
Political dependence on hugely expensive nuclear capability. Wants to increase but probably has no more than one or two warheads operational. Intentions depend on present six-power talks to scrap nuclear weapons programme

GetBulb



Data visualisation

- By contrast, the term data visualisation is useful for referring to any visual representation of data that is:
 - **Algorithmically drawn**: may have custom touches but is largely rendered with the help of computerized methods
 - **Easy to regenerate with different data**: the same form may be repurposed to represent different datasets with similar dimensions or characteristics
 - **Maybe aesthetically barren**: data is not decorated
 - **Relatively data-rich**: large volumes of data are welcome and viable, in contrast to infographics

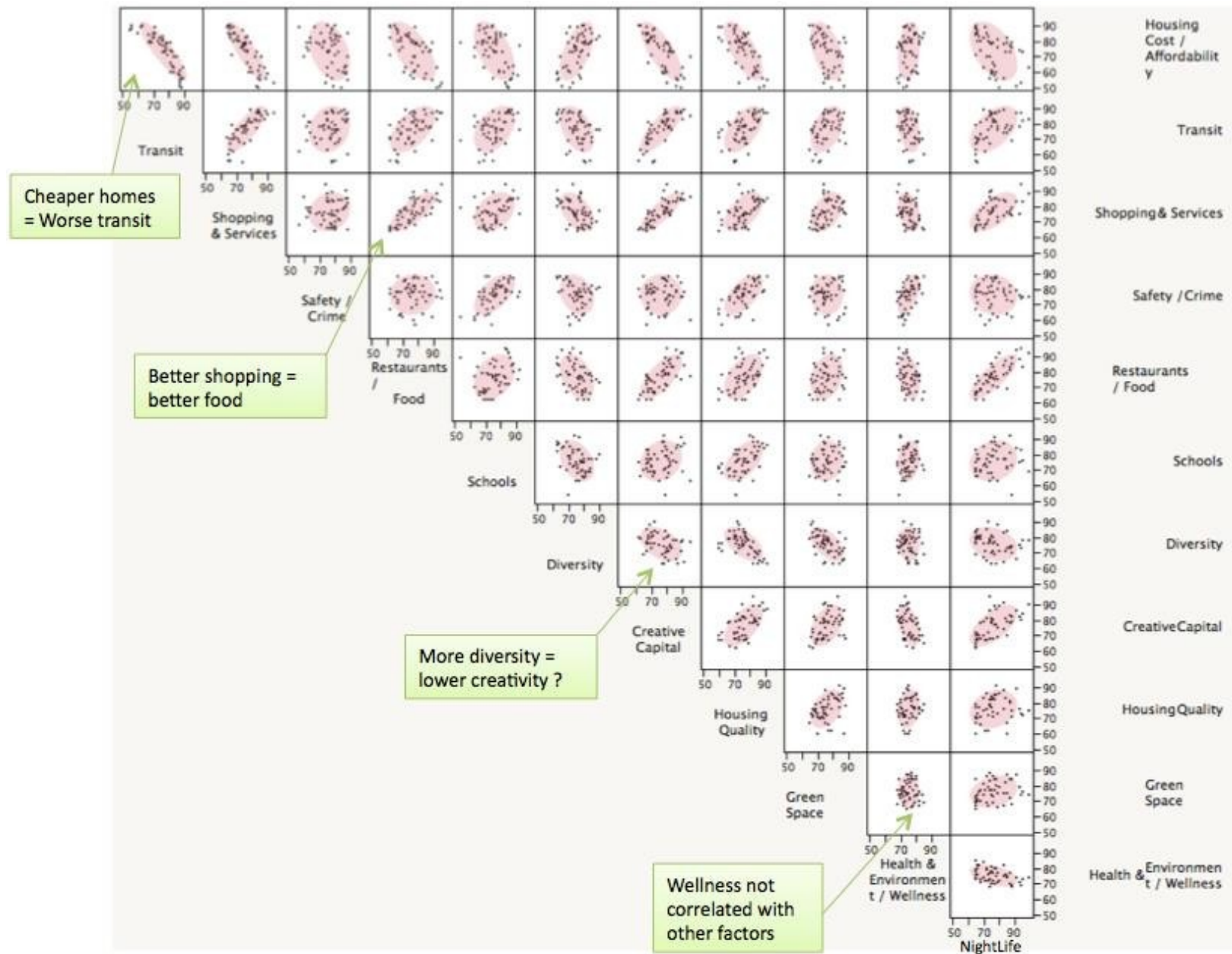


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Exploration

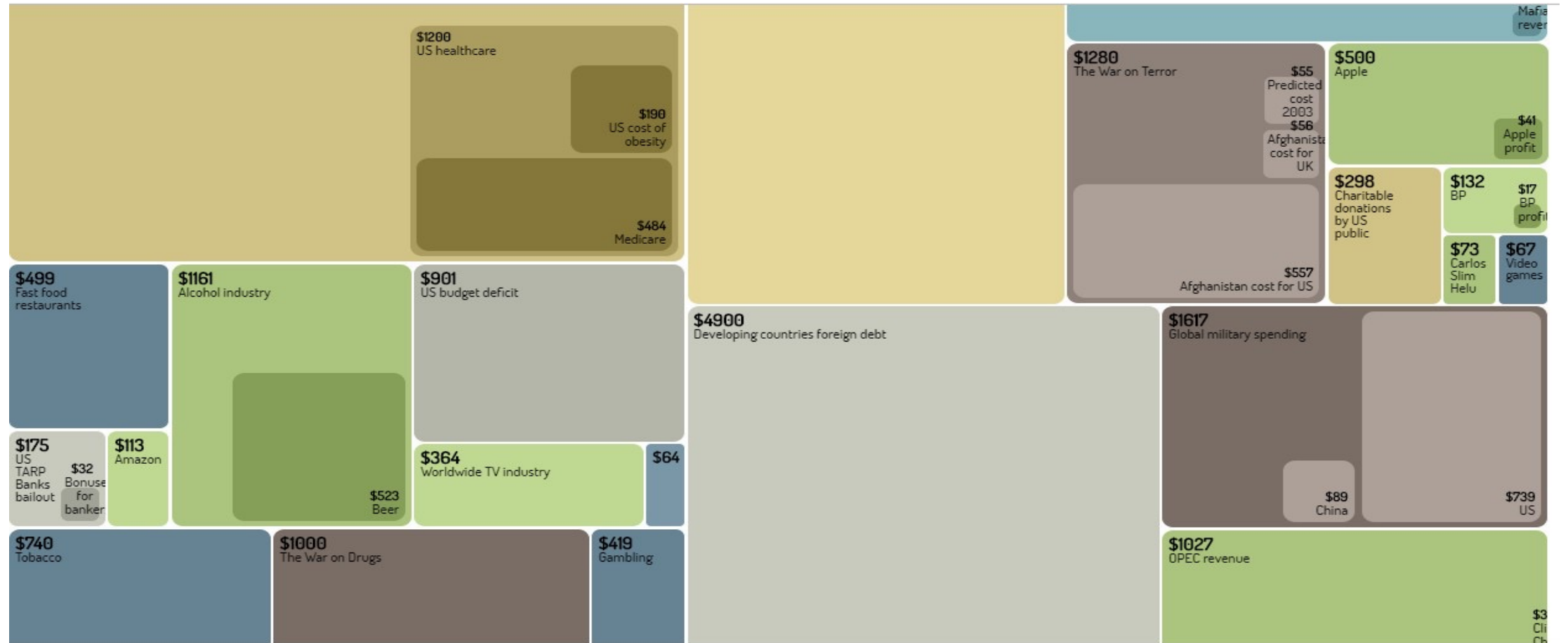
- Exploratory data visualisations are appropriate when you have lots of data and you are not sure what's in it
 - When you need to get a sense of what's inside your data set, translating it into a visual medium can help you quickly identify its features, including interesting curves, lines, trends, or anomalous outliers
- This type of visualisation is typically part of the data analysis phase, and is used to find the story the data has to tell you



Explanation

- Explanatory data visualisation is appropriate when you already know what the data has to say, and you are trying to tell that story to somebody else
 - Whoever your audience is, the story you are trying to tell is known to you at the outset, and therefore you can design to specifically accommodate and highlight that story
- Explanatory data visualisation is essentially part of a presentation phase - classic slideware!

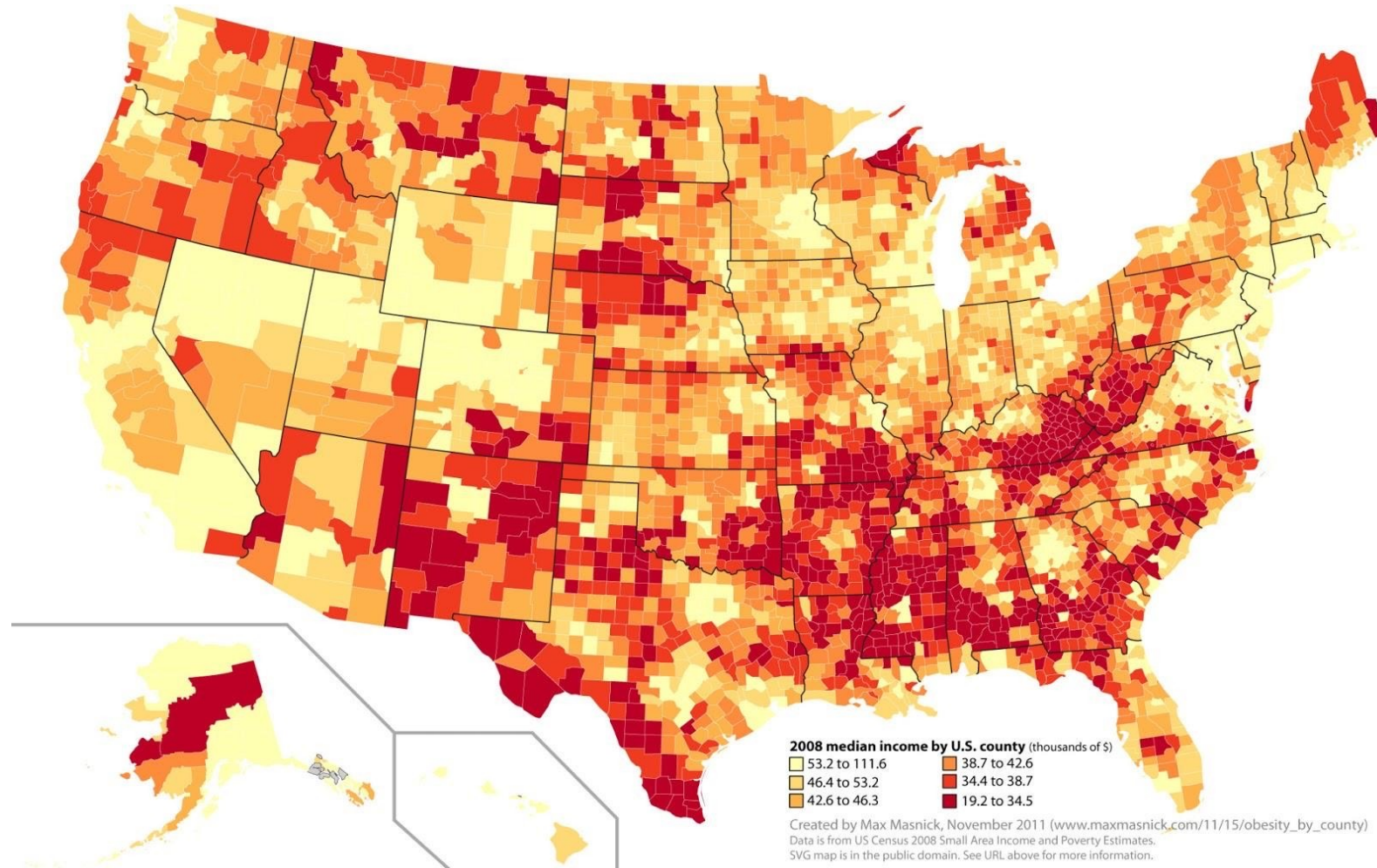
Billion-dollar-o-gram



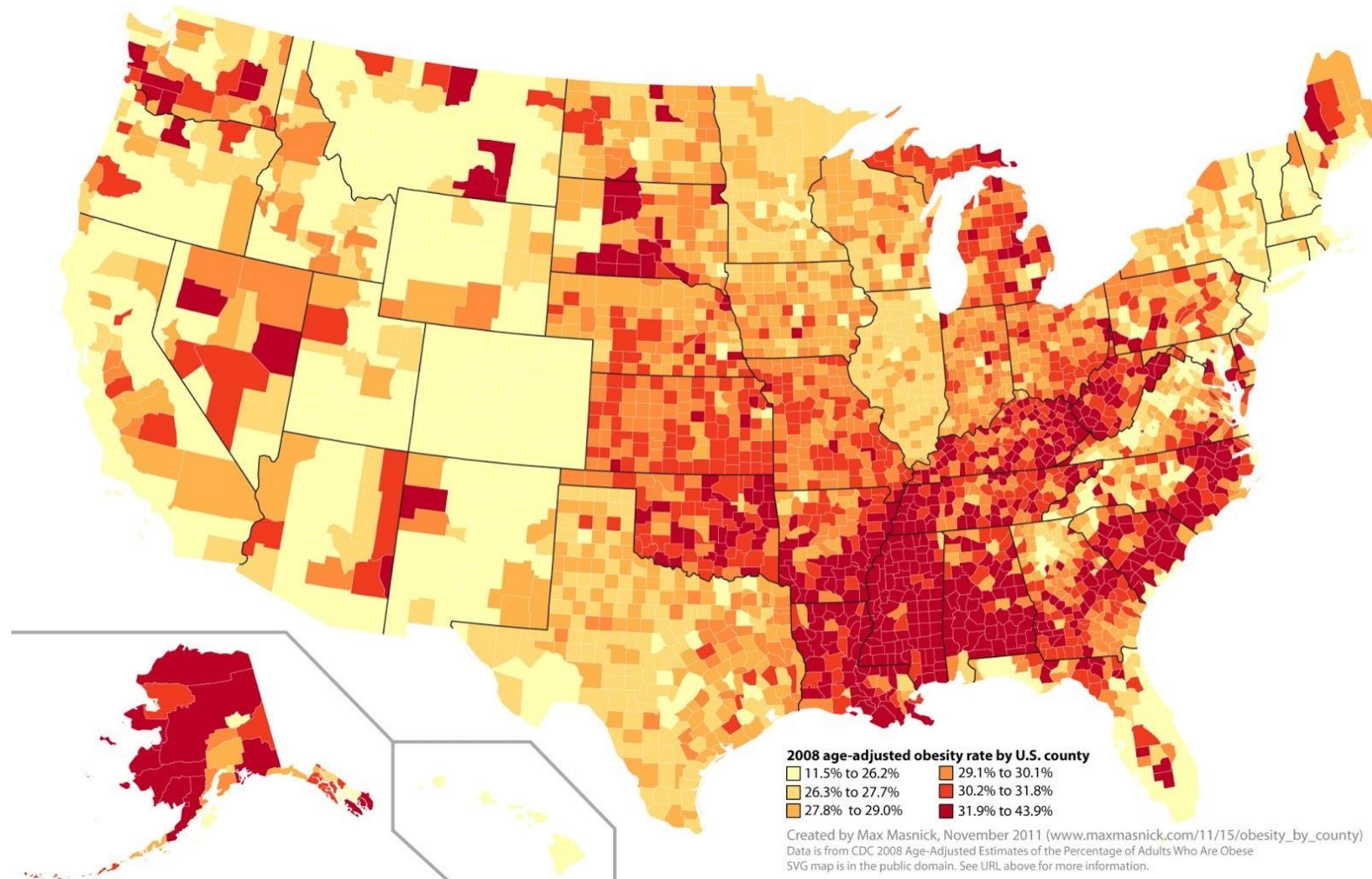
Hybrids: Exploratory & Explanatory

- Hybrid category: a curated dataset that is nonetheless presented with the intention to allow some exploration on the reader's part
 - These visualisations are usually interactive via some kind of graphical interface that lets the reader choose and constrain certain parameters
 - Sometimes allows the reader discover insights the designer of the visualisation hasn't come across yet!
- In these hybrid designs there is a certain freedom- of-discovery aspect to the information presented, but it is usually not totally raw; it has been distilled and facilitated in some way

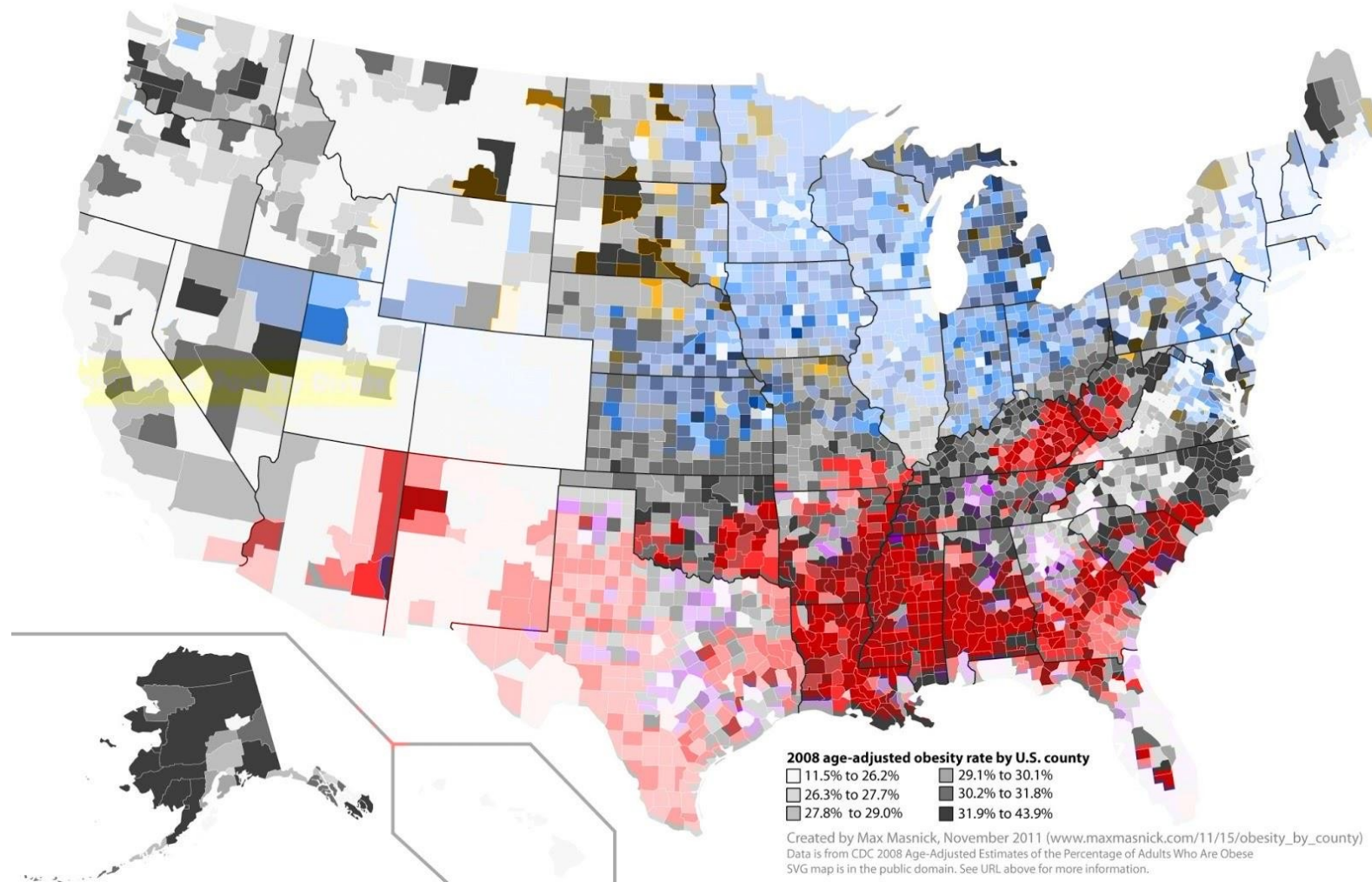
Income Map USA



Obesity Map USA



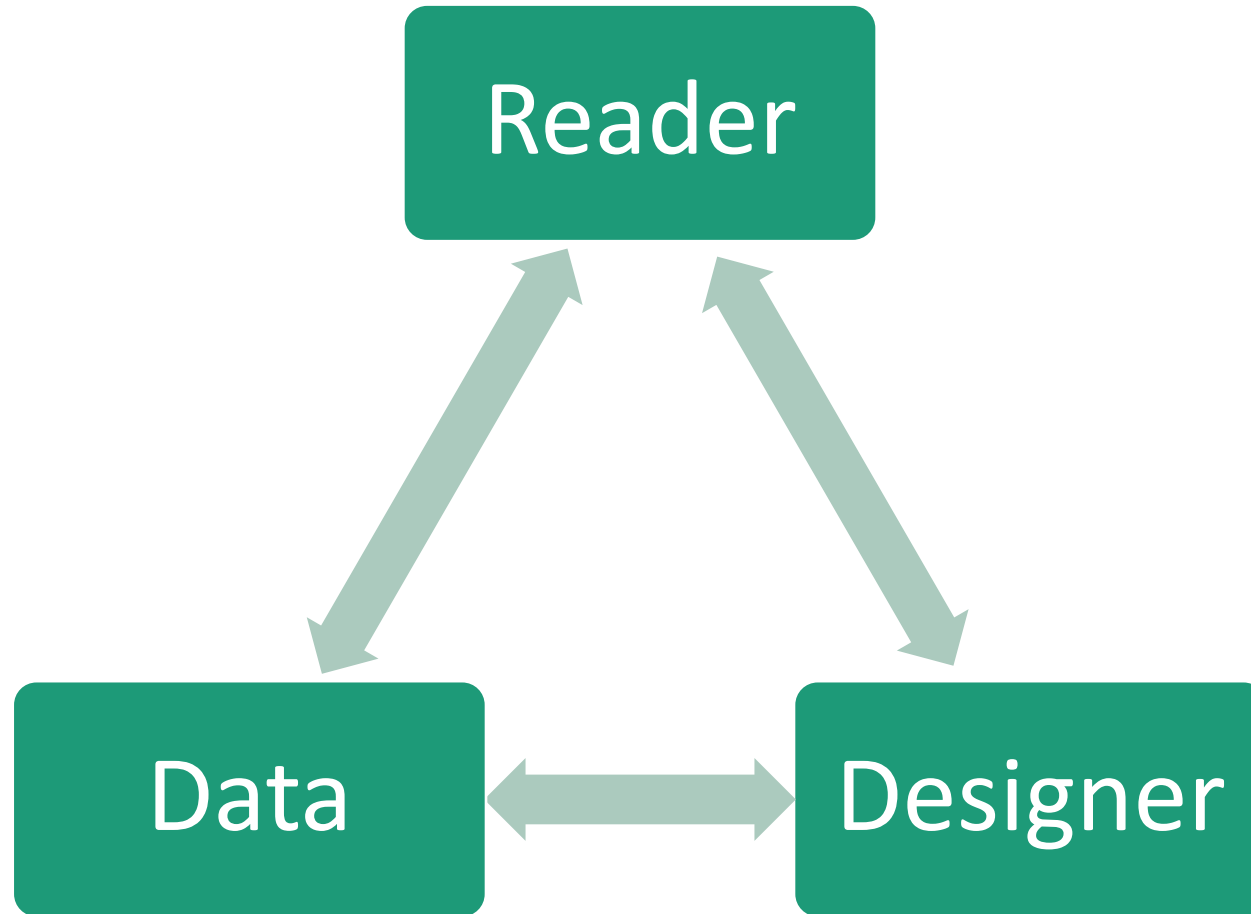
Correlated Map



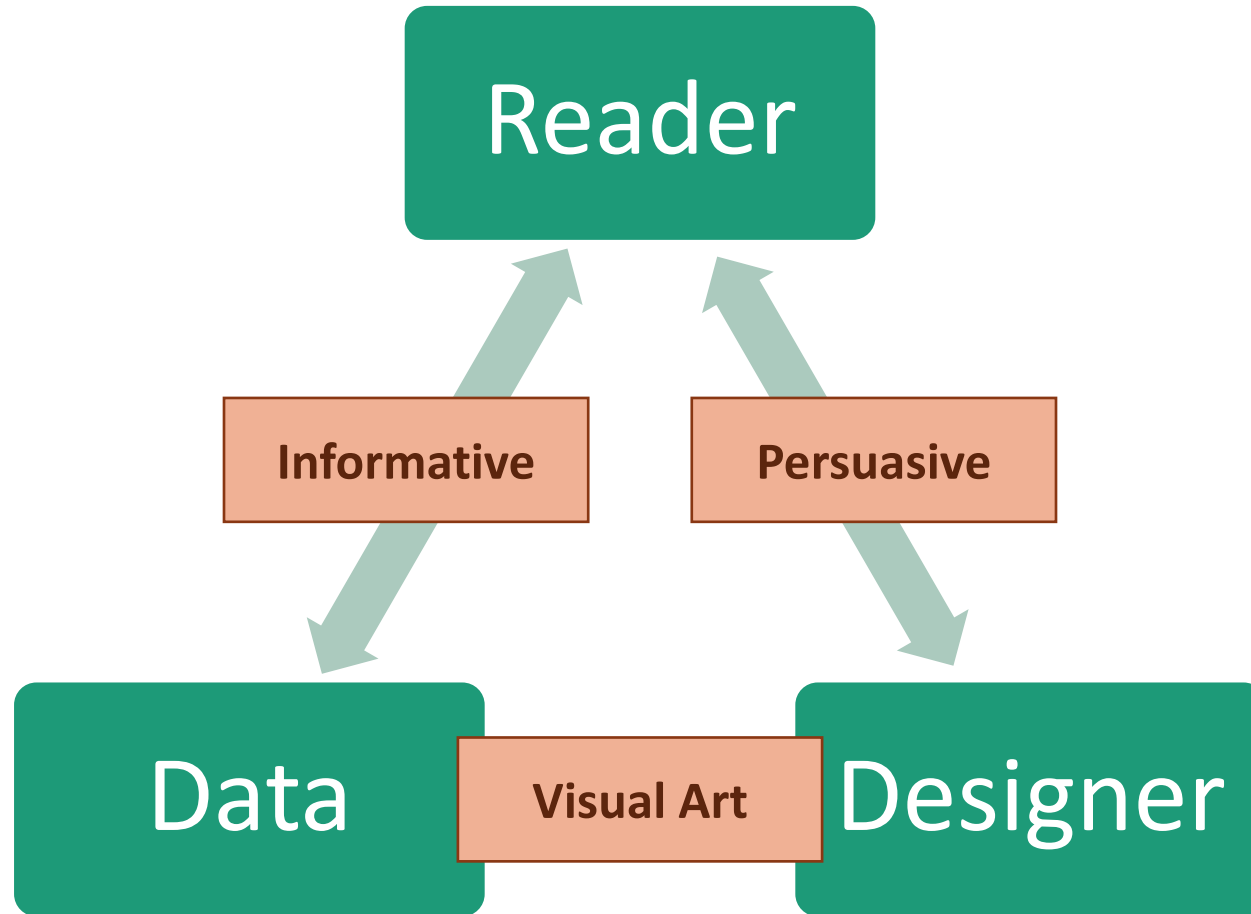
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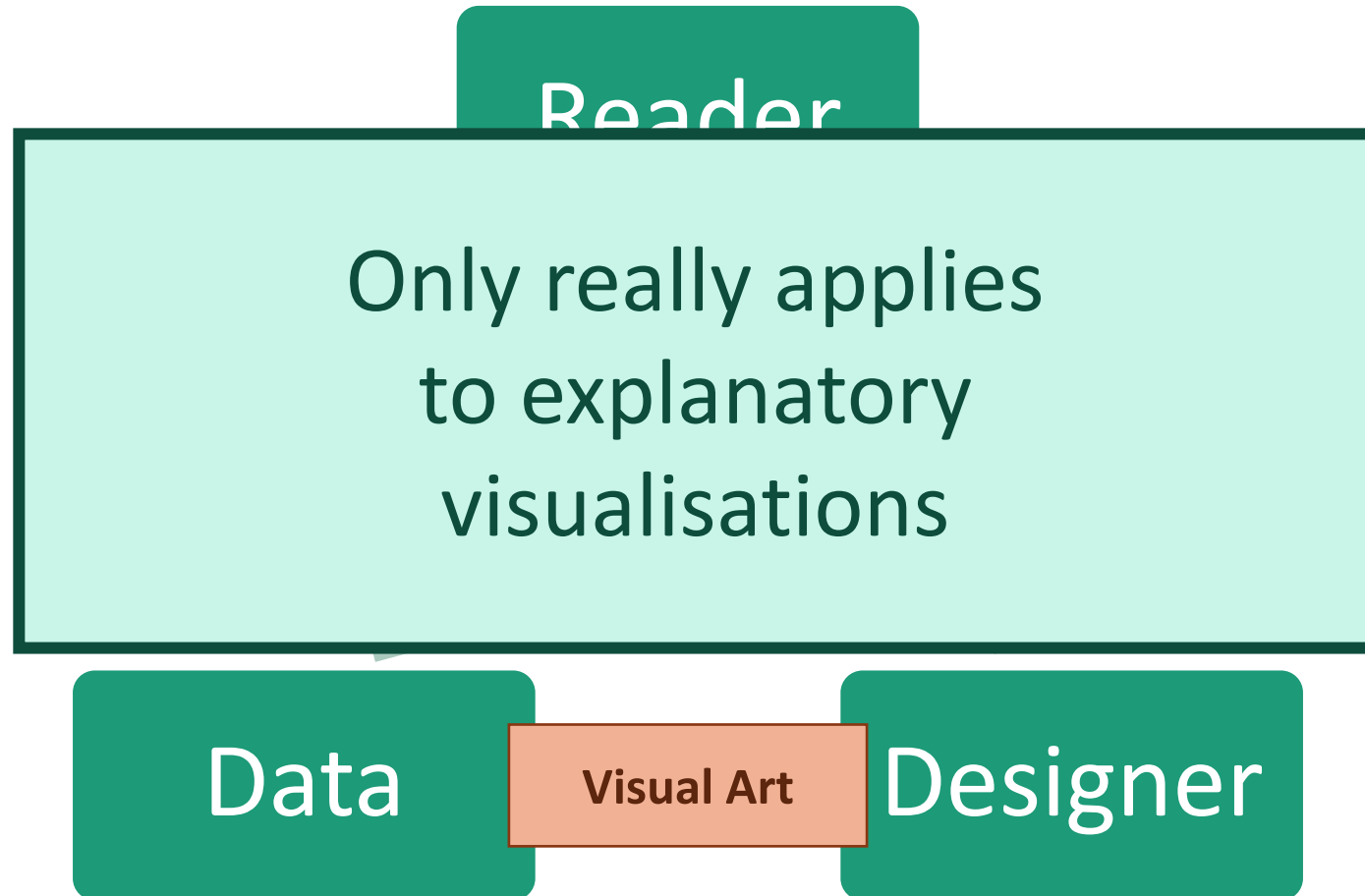
The Designer-Reader-Data Trinity



The Designer-Reader-Data Trinity



The Designer-Reader-Data Trinity



The Designer-Reader-Data Trinity

- We can think of explanatory data visualisation as being supported by a three-legged stool consisting of the designer, the reader, and the data
- Each of these "legs" exerts a force, or contributes a separate perspective, that must be taken into consideration for a visualisation to be stable and successful
- Each of the three legs of the stool has a unique relationship to the other two
- While it is necessary to account for the needs and perspective of all three in each visualisation project, the dominant relationship will ultimately determine which category of visualisation is needed

Informative

- **An informative visualisation primarily serves the relationship between the reader and the data.**
- It aims for a **neutral presentation** of the facts in such a way that will educate the reader (though not necessarily persuade him).
- Informative visualisations are often associated with broad data sets, and seek to distil the content into a manageably consumable form.

Google's Share Price



Persuasive

- A persuasive visualisation primarily serves the relationship between the designer and the reader.
- **It is useful when the designer wishes to change the reader's mind about something.**
- In this category of visualisation, the data represented is specifically chosen for the purpose of supporting the designer's point of view, and is presented carefully so as to convince the reader of same.

Persuasive

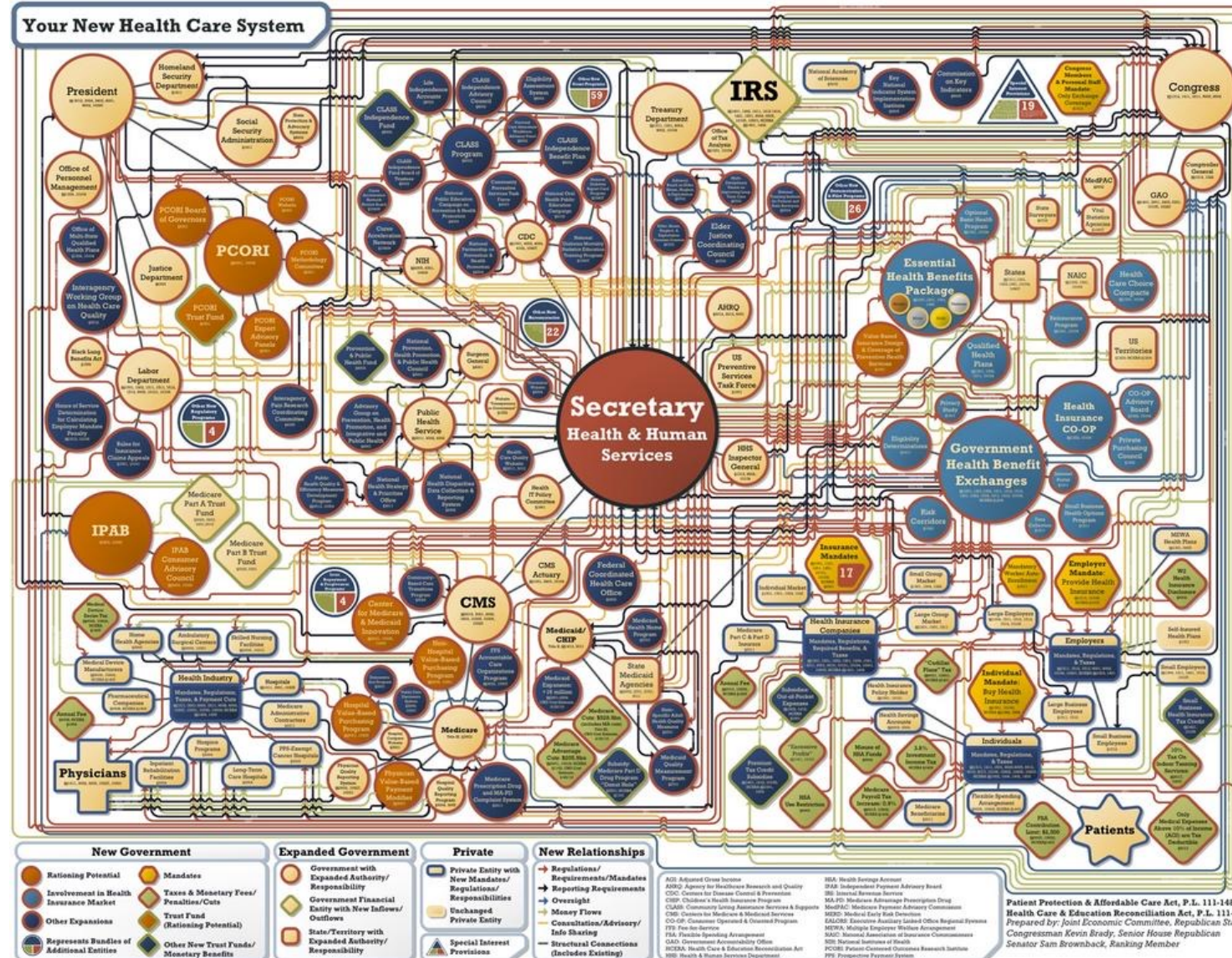
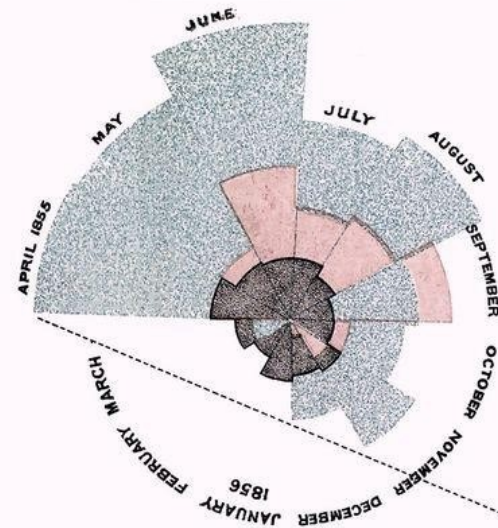
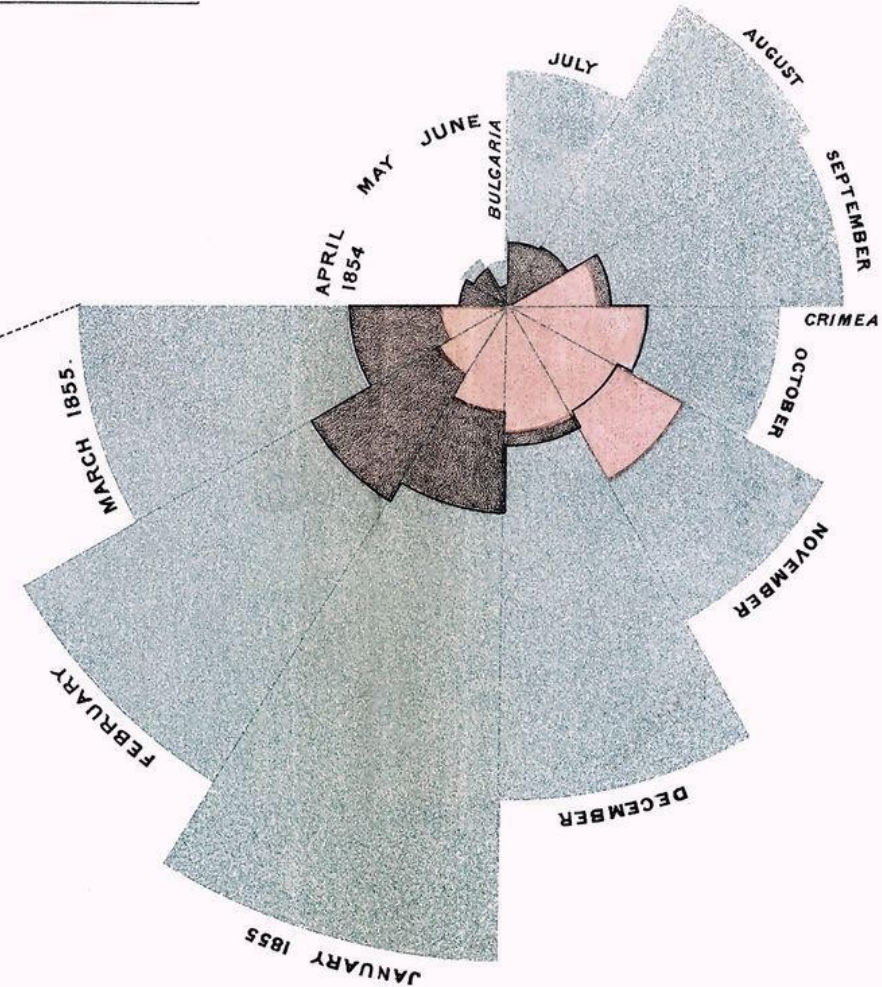


DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.

2.
APRIL 1855 TO MARCH 1856.



1.
APRIL 1854 TO MARCH 1855.



The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.

The blue wedges measured from the centre of the circle represent area for area the deaths from Preventable or Mitigable Zymotic diseases; the red wedges measured from the centre the deaths from wounds; & the black wedges measured from the centre the deaths from all other causes.

The black line across the red triangle in Nov^r 1854 marks the boundary of the deaths from all other causes during the month.

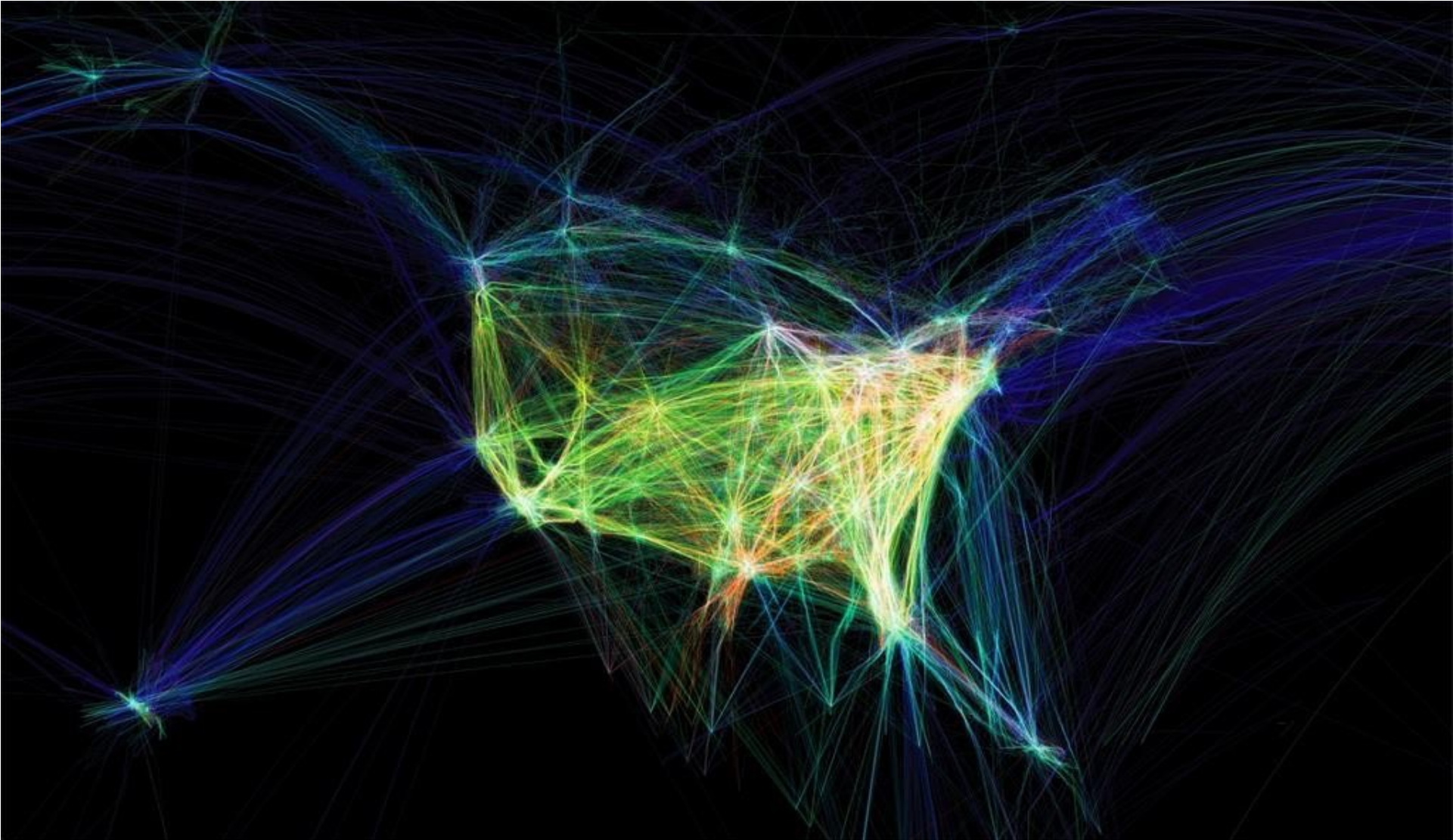
In October 1854, & April 1855, the black area coincides with the red; in January & February 1856, the blue coincides with the black.

The entire areas may be compared by following the blue, the red & the black lines enclosing them.

Visual Art

- The third category, visual art, primarily serves the relationship between the designer and the data.
- Visual art is unlike the previous two categories in that it often entails **unidirectional encoding of information, meaning that the reader may not be able to decode the visual presentation to understand the underlying information.**
- The designer may intend only to condense it, translate it into a new medium, or make it beautiful; she may not intend for the reader to be able to extract anything from it other than enjoyment.

Flight Patterns, Aaron Koblin



Facebook Visualised



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Classifications of visualisations

- There are many different ways to classify visualisations
- Classifications can be useful when discussing/planning visualisation projects
- Classifications are also useful when determining the appropriate visualisation techniques to deploy

Interesting Reading

- Open Mind Blog.
 - USA Temperature: can I sucker you?
 - Link: <https://tamino.wordpress.com/2018/08/08/usa-temperature-can-i-sucker-you/>

Video

- Weather visualisation
 - https://youtu.be/q01vSb_B1o0?t=46

Thanks To

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